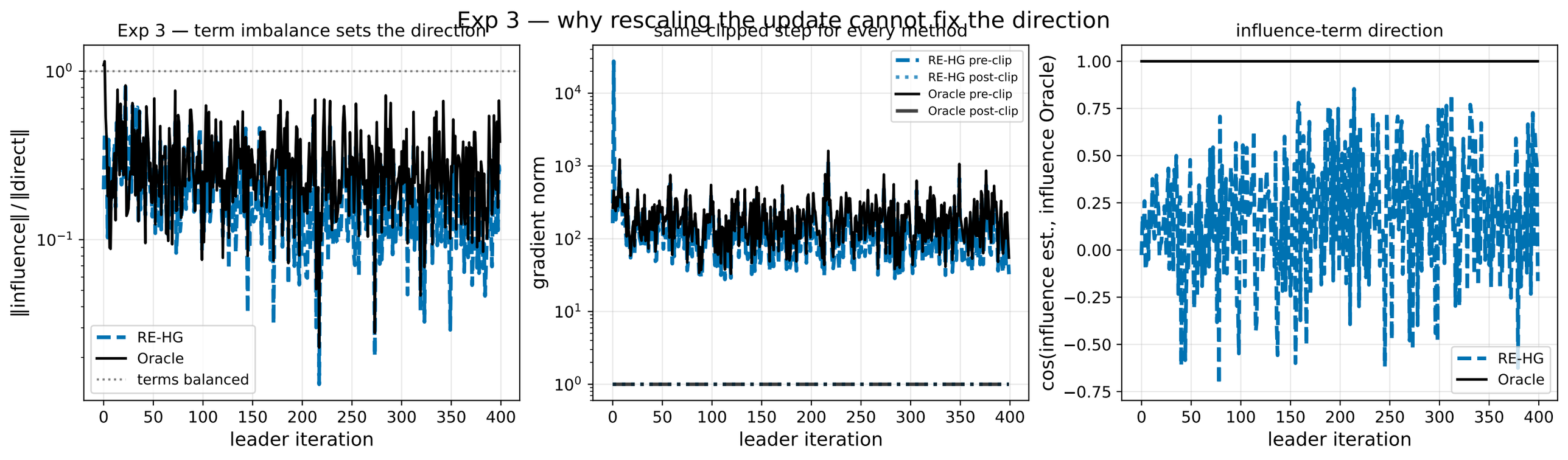
Exp 3 — why rescaling the update cannot fix the direction
Exp 3 — term imbalance sets the direction
‖influence‖ / ‖direct‖
10^0
10^{-1}
RE-HG
Oracle
terms balanced
0
50
100
150
200
250
300
350
400
leader iteration
same clipped step for every method
gradient norm
10^4
10^3
10^2
10^1
10^0
RE-HG pre-clip
RE-HG post-clip
Oracle pre-clip
Oracle post-clip
leader iteration
influence-term direction
cos(influence est., influence Oracle)
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
RE-HG
Oracle
leader iteration